Quality goals and design principles

HCI Design, evaluation and implementation of interactive computing systems for human use, and study of the human phenomena around them. Judge effects for specified people, goals and contexts.

Usability / UX Usability concerns effectiveness, efficiency and satisfaction in context. UX also includes trust, pleasure, frustration, engagement and meaning. Success, time and subjective reactions provide different evidence.

Utility / usefulness Utility asks whether the system has the functions people need. Usefulness requires both utility and sufficient usability. Fast interaction cannot compensate for a missing core function.

Six usability goals Effectiveness: correct and complete. Efficiency: time and resources. Safety: prevent errors and limit harm. Utility: suitable functionality. Learnability: first use. Memorability: returning after a gap. Fast but error-prone can be efficient yet ineffective.

Goal → measure Specify user, task, context and target. Success rate measures effectiveness. Time and effort measure efficiency. Error consequences and recovery inform safety. First-use and delayed tests distinguish learnability and memorability.

Authority and generality Standard: formal specification. Guideline: practical advice. Principle: abstract idea. Pattern: reusable solution in context. Anti-pattern: recurring poor solution. Dark pattern: manipulates or deceives users.

Six design principles Visibility makes actions and states discoverable. Feedback makes results interpretable. Consistency supports transfer. Affordance is a possible action. Constraint limits invalid actions. Mapping links control and effect. A signifier indicates a possible action.

How models reach users Designer conceptual model → system image (interface, feedback, documentation) → user mental model. A metaphor can convey the model. Feedback can correct mistaken expectations.

Two gulfs Execution: how do I do it? Improve entry points, labels or mappings. Evaluation: what happened? Clarify status, progress and recovery. A walkthrough applies these questions step by step.

Nielsen's 10 System status visibility, match with the real world, user control and freedom, consistency and standards, error prevention, recognition over recall, flexibility and efficiency, aesthetic minimalism, error recognition/diagnosis/recovery, help and documentation. Keep task-critical information.

Shneiderman's 8 Consistency, shortcuts, informative feedback, dialog closure, error prevention and simple handling, easy reversal, sense of control, low short-term memory load. Closure means a clear end to a sequence.

Perception, cognition and movement

Information processing Sensory input → attention, memory and judgement → action → system output. Foveal vision reads detail. Peripheral vision detects salient change. Size, position, space and contrast establish hierarchy.

Seven Gestalt laws Proximity groups nearby items. Similarity groups alike items. Common fate groups items moving together. Continuity follows a path. Closure fills gaps. Symmetry groups mirrored forms. Figure/ground separates object from background.

Sound and touch Intensity relates to loudness, frequency to pitch, timbre to sound quality. Alerts attract attention but can interrupt and convey precise values poorly. Tactile sense is skin-based. Kinaesthesia concerns body position and motion.

Mechanoreceptors Merkel: pressure, 0–30 Hz. Ruffini: stretch, 0–15 Hz. Meissner: local vibration, 10–60 Hz. Pacinian: less local vibration, 80–400 Hz. Approximate lecture ranges. Fingertip discrimination thresholds do not specify target spacing.

Memory Sensory: brief. Working: holds and manipulates, with executive, visuospatial and articulatory components. Long-term: semantic, procedural and episodic. Recognition uses cues. Recall requires retrieval. Show choices, retain input and chunk meaningfully. 7 ± 2 is a teaching approximation.

Attention and load Selective attention focuses. Divided attention shares limited resources. Stroop: reading interferes with naming ink colour. External cognition offloads work onto representations. Distributed cognition spans people, tools and time.

Fitts' Law $MT = a + b \log_2(1 + D/W)$, with distance D, effective width W along approach, and fitted a, b. Smaller D/W predicts faster movement. Equal ratios predict equal times with the same parameters. Search, choice and reaction need separate analysis.

Fitts comparison At $D = 160$, $W = 20$, difficulty $ID = \log_2 9$. Doubling width gives $ID = \log_2 5$. Different label difficulty can still change total task time.

Human capabilities → interface choices

Interaction / interface type Instructing gives commands. Conversing exchanges turns and clarifies. Manipulating acts on objects. Exploring navigates space. Responding answers a system prompt. Interface type is the form or medium. Speech can carry commands or conversation.

Direct manipulation Continuous representation of objects/actions, rapid reversible incremental operations and immediate feedback, and direct physical action in place of complex syntax. A touchscreen can still trigger a hidden command.

CLI / GUI CLI supports precision, scripts and batch tasks at the cost of syntax recall. GUI offers visible cues for exploration. WIMP = Windows, Icons, Menus, Pointing device.

Post-WIMP / reality-based Uses at least one technique beyond traditional 2D widgets. GUI elements can coexist. Reality-based interaction draws on naive physics, body skills, environmental skills and social skills. Familiar movements still need clear mappings.

Choose a modality Speech is hands-free but sensitive to noise, accents, privacy and repair. Interactive surfaces support multitouch or sharing. Check reach, occlusion and conflicts. Tangible interfaces use physical objects to represent and control digital state. Check synchronisation.

Space and body VR provides an interactive simulated environment. Check fatigue and safety. Wearables support mobility and context sensing. Check comfort, battery and acceptability. BCI links brain signals to devices. Check training, reliability and feedback.

Three accessibility checks Is foreground readable against background? Can people with different colour vision distinguish states? Is meaning available without hue? Contrast tools, colour-vision simulation and task testing answer different questions.

Redundant coding Pair meaningful colour with text, shape, icons, line style or position. Keep meaning consistent and check cultural expectations. Red and green lines can each contrast with white yet still need labels or line styles.

Legibility Small blue text or thin blue lines on dark backgrounds can be hard to read. Check actual luminance, size, stroke and lighting. Test focus, selected, disabled and error states, plus keyboard paths and real tasks.

SIID / PACT Situationally induced impairments and disabilities can arise from noise, cold, glare, loads, walking or divided attention. Try larger targets, one-handed paths, pausable input, reviewable feedback or alternate modalities. Test the context effect, then the adaptation.

Social interaction and emotion

Turn-taking The speaker selects the next speaker, someone self-selects, or the current speaker continues. Gaze, posture and tone signal turns. Delay and missing cues can call for hand raising, active-speaker cues, clarification or replay.

Presence Telepresence enables remote participation, e.g. via a robot. Social presence is the felt presence and responsiveness of others online. Co-presence concerns shared physical space and activity.

Awareness / coordination Awareness is knowing who is present and what is happening. Coordination organises joint action via rules, schedules and artefacts. Live cursors aid awareness. Edit locks aid coordination. Peripheral awareness is incidental. Situational awareness includes implications.

CSCW time/place matrix Same place/same time: shared whiteboard. Same place/different time: handover board. Different place/same time: video call or live co-editing. Different place/different time: email or forum. Classify the activity. One tool can span cells.

Emotional design Visceral: first impression. Behavioural: use, function and performance. Reflective: meaning, identity and value. Maslow frames motivation as physiological, safety, belonging, esteem and self-actualisation needs.

Valence / arousal Pleasantness and activation are separate axes. Anger and excitement can both have high arousal. Intended elicitation, SAM self-report and sensor readings are distinct. Consider order and lingering emotion.

Understand → sense → adapt Establish a context effect, validate sensing, then test adaptation. Face, EDA, speech, movement and task metrics are cues. Interpret them with context and individual baselines.

Diagnose coordination Unclear who claimed an item suggests missing awareness. Duplicate claims suggest missing coordination. Show relevant status while limiting exposure.

Choose a fallback In a noisy workshop, speech can fail. Gloves can also impair small touch targets. Identify the affected channel before comparing larger targets, physical controls, text or haptics.

Design process → requirements

Design orientations User-centred: user needs and goals. Activity-centred: activities. Systems design: components and relations. Genius design: designer judgement. User involvement can manage expectations and improve ownership. Designers still make trade-offs.

Three UCD principles Early focus on users and tasks, empirical measurement, iterative design. A late preference survey alone is insufficient. Revisit requirements, alternatives, prototypes and evaluation as evidence changes.

Double Diamond Discover diverges to explore the problem. Define converges on a challenge. Develop diverges across ideas and prototypes. Deliver converges on a refined solution. Problem space first, solution space second.

AgileUX Short cycles deliver working increments and reprioritise with feedback. Iteration 0 establishes needed research and vision. Design can run one or two iterations ahead. Minimal documentation supports collaboration and traceability.

PACT People: abilities, experience, goals. Activities: frequency, risk, interruptions and collaboration. Contexts: physical, social, organisational. Technologies: input, output and communication. Look for interactions, e.g. gloves with small touch targets.

Stakeholders Operators, people affected indirectly, purchasers and administrators may have conflicting goals. Ask how work happens, why people bypass rules and what information they need. A requested feature can point to a deeper goal.

Six requirements Functional: actions. Data: content, quality, updates. Environment/context: use conditions. User: people, skills, use patterns. Usability: performance. UX: intended experience. Trace each requirement to evidence and plan a check.

MoSCoW Must is essential for a viable release. Should matters but the system works without it. Could adds value and can wait. Won't is explicitly out this round. Prioritise by goals, risk and resources.

Research question → evidence

Before collecting data Define goals and analysis, recruit suitable participants, manage relationships and consent, triangulate sources, pilot questions and procedures. Investigate circumstances when sources disagree.

Survey / questionnaire A survey includes sampling, recruitment, instrument and analysis. A questionnaire is the set of questions. Useful for attitudes, intentions and distributions. Self-reported use can have recall bias.

Sampling Probability sampling has known inclusion probabilities, e.g. random or stratified random. A census covers everyone. Snowball, self-selection and volunteering are non-probability. Large samples can still be biased. Sampling selects people. Allocation assigns conditions.

Question design Open questions explore reasons but take work to code. Closed questions compare readily but may omit options. Distinguish ordered and unordered responses. Label Likert anchors and direction. Avoid leading, double-barrelled or double-negative items and vague time frames.

Interviews Unstructured explores deeply. Semi-structured shares core prompts with probes. Structured fixes wording and order. Introduce, warm up, ask main questions, cool down and close. Probe concrete events neutrally to reduce approval bias.

Focus groups Interaction reveals norms and disagreements. Balance dominant and quiet members. Power or sensitive topics can suppress disclosure. Shared agreement does not yield independent observations.

Observation / logging Direct observation can be silent, think-aloud or paired constructive interaction. Speaking can change task pace. Indirect sources include video, logs and diaries. Record actions and times separately from interpretation.

Diaries / documents Event- or time-based diaries capture dispersed, private or longitudinal activity. Feedback diaries report directly. Elicitation diaries prompt interviews. Documents describe prescribed work. Observe how work actually happens.

Contextual inquiry / ethnography Inquiry combines on-site observation and interview using context, partnership, interpretation and focus. Ethnography studies a group's culture and practices, often analysing during collection. Brief field observation alone is insufficient.

Recording trade-offs Notes are flexible but can omit detail. Audio preserves wording, photos preserve space, video preserves sequences with greater analysis and privacy costs. Choose by question, participants, skills and resources.

Observation frame Who, where and what. Expand to space, actors, activities, objects, acts, events, time, goals and feelings. A 12-second pause is observable. Attributing it to fear needs corroboration.

Raw data → interpretation and task models

Descriptive statistics Check missing data, duplicates, outliers and logical conflicts. Retain originals and justify exclusions. Category codes do not imply numeric distance. Mean $\bar{x} = \sum x_i / n$ is sensitive to extremes. Median is the middle, mode the most common, SD measures spread.

Qualitative evidence A critical incident matters for its impact, even if rare. Inductive analysis derives themes. Deductive applies predefined categories. Affinity diagrams cluster observations. Preserve provenance, counterexamples and context.

Analysis frameworks Conversation: turns, pauses and intonation. Discourse: how language constructs meaning. Content: coded media features. Interaction: verbal and nonverbal activity. Grounded theory: theory from data. Systems-based: people, organisations and technology.

Profile / persona / scenario A profile summarises attributes. A persona synthesises evidence into a representative character and goals. A scenario describes who acts, where, what triggers action and how it unfolds. It can inform use cases and test tasks.

HTA Root goal typically numbered 0. The hierarchy decomposes tasks. A plan combines subtasks in sequence, any order, parallel, branching, loops, or selection of alternatives. Choose depth to fit the analysis goal.

GOMS Goals, Operators, Methods, Selection rules. An operator is a basic action, a method a set of steps, and a selection rule chooses the method. HTA emphasises decomposition and plans. GOMS emphasises operations and strategy.

Use cases Actors can be people, organisations or external systems. Ovals represent use cases, links show participation, and the boundary shows scope. Essential use cases pair user intentions with system responsibilities. Detailed cases specify main, alternate and exception flows.

Task model → concepts and prototypes

Conceptual model Define objects, operations, relations, function grouping and order, information and responsibilities. Extract domain nouns, attributes and verbs from scenarios. Menus, buttons and buffers are interface or implementation details.

Metaphor Use familiar experience to explain a model. Compare relevant structure, ease of representation, familiarity and extensibility. Digital interactions can go beyond the physical analogue.

Representations Sketch: quick alternatives. Storyboard: people, setting and events. Wireframe: one screen's layout. Wireflow: screens, actions and branches. Prototype: interaction users can try.

Scope and fidelity Horizontal: broad and shallow, useful for structure and findability. Vertical: narrow and deep, useful for a critical path. Fidelity concerns resemblance in look, content and interaction. Digital can still be low fidelity.

Pick a prototype Paper is easy to revise. The operator changes screens neutrally. Clickthrough links hotspots, so a dead end may be a prototype limit. Physical prototypes explore tangible or wearable interaction. Video conveys a scenario. Scripted success does not establish usability.

Wizard of Oz A human behind the scenes simulates unfinished speech, intelligence or automation. Standardise responses and exceptions. Plan ethics. Simulated response times do not establish software performance.

Evidence to validation Observe a workaround → probe reasons → formulate a measurable requirement → use HTA to identify exceptions → prototype → inspect discoverability → test independent completion with target users.

Apply the models

HTA plan example 0 shop, 1 confirm need, 2 prepare list, 3 buy, 4 sync. Plan 0 might be 1–2–3–4. Plan 3 repeats check, choose and mark for each item, returning to choice if out of stock. Numbering locates tasks. The plan specifies execution.

Essential use case example User intention: confirm my household can see it. System responsibility: report sharing scope and sync outcome. Button choice belongs in concrete design.

From evidence to measures Duplicate shopping could reflect hidden status, unclear responsibility or failed sync. Test whether users can identify purchase status and responsibility. Measure accuracy and time separately.

Prototype inference Test app-wide categories with broad navigation. Test failed submission recovery with a deep exception path. Separate design failures from unimplemented hotspots.

Research goal → evaluation method

When and where Formative during iteration, acceptance before release, logs and field studies after release. Lab offers control. Field offers realism. Remote describes participant/researcher separation and can support either.

Pick the method Principle violations: heuristic inspection. First-use discovery: walkthrough. Target-user performance: usability test. Hypothesis about a manipulated factor: experiment. Natural or longitudinal use: field. Behaviour at scale: analytics. Performance prediction: model.

Heuristic evaluation Brief experts → inspect independently, first overall then in detail → consolidate and rank. Record location, trigger, principle, impact and fix. Inspection can miss issues or raise false alarms. Five experts finding 75–80% is a rule of thumb.

Severity 0–4 0 no problem, 1 cosmetic, 2 minor, 3 major, 4 catastrophe requiring a fix before release. Consider frequency, impact and recovery. A rare irreversible error can outrank a common cosmetic issue.

Cognitive walkthrough Define novice knowledge, task and correct sequence. At each step: will users form the right subgoal, see the action, connect it to their goal, and interpret feedback as progress? Record predicted breakdowns and reasons.

Analytics / A/B / model Logs show events. Long dwell time could mean engagement or confusion. A/B compares real users and needs comparable conditions, allocation, clear metrics and privacy safeguards. Fitts predicts one performance component within model assumptions.

Field / in-the-wild Study actual use, adoption and change in natural settings. Combine observation, event reports, prompts and logs. Consider reactivity, household privacy, consent, failures and safety. Realism alone does not establish causality.

Run a usability test

Who and what Recruit target users. Cover representative roles and frequent or critical tasks. Give the goal and context without click instructions. Standardise wording, start state, equipment and assistance.

Operationalise measures Define independent success, assisted success, partial success and failure in advance. Set timing boundaries and what counts as an error or help. Record time, errors, incidents and satisfaction. Success-only times can hide failures.

Five team roles Facilitator runs sessions neutrally. Prototype executor simulates responses. Quantitative collector records time/errors. Qualitative collector records behaviour/incidents. Supporting actors play roles and manage props.

Sample and pilot Three to five users is a discovery rule of thumb. Lectures also suggest 5–10 or stopping when new insights cease. Multiple roles, complex tasks and comparisons require separate planning. Pilot hotspots, duration and recording rules.

Consent and session Participant Information Statement (PIS) explains procedures, data, risks and rights. Participant Consent Form (PCF) records informed consent. NDA protects confidentiality. Screen → observe tasks → probe, collect post-test measures (e.g. SUS, USE), debrief. Plan withdrawal, access and retention.

Test / experiment Usability testing seeks product problems and improvements, often with small iterative samples. Experiments test variable relationships with greater control, replication and sample planning. Bound conclusions by studied users, tasks and conditions.

Make findings auditable

Example coding rule If a participant saves correctly but sync fails, use pre-set criteria. If they succeed after a prompt, record the assistance. Report failures alongside timing summaries.

Problem report Task and start state → action, state and consequence → mechanism or violated principle → fix → retest measure. Justify severity with evidence.

Ticket-machine walkthrough No feedback after destination selection signals an evaluation gulf. Missing cues to choose trip type signal an execution gulf. Show amount paid, remaining balance and cancellation. Cover insufficient payment.

Priority example A misaligned decoration and irreversible deletion can both be observable. Recovery cost and consequence make the deletion more urgent even if rare.

Triangulate an anomaly When many users open help, observe where they get stuck and interview them about why. Logs alone cannot distinguish confusion from purposeful reference.

Ethics throughout Explain and pilot first. During sessions, avoid pressure and judgement, offer breaks and allow withdrawal. Debrief and handle data as agreed. Approval and participant consent serve different roles.

Experimental design → statistical inference

Hypothesis, variables, control A hypothesis is a testable prediction. IV is a factor with levels. DV is an outcome. Control is the comparison baseline. Age, gender or experience can be grouping attributes rather than manipulable factors.

Between-subjects Different participants per condition avoid cross-condition learning and fatigue. Individual differences can obscure effects. Randomise allocation, standardise procedures and plan sample size. Representative sampling is a separate question.

Within-subjects The same people do each condition. Learning, fatigue and carryover can distort results. Counterbalance AB/BA and match task difficulty. Everyone doing A first confounds condition with order.

Matched participants Match on relevant attributes and assign members of each pair to different conditions. Matching costs effort and controls only measured, matched differences.

One factor, two levels Independent groups → independent-samples t test. Same people twice → paired-samples t test. Identify dependence and check assumptions.

One factor, three or more levels Independent groups → one-way ANOVA. Repeated observations of the same people → repeated-measures ANOVA. Many levels can still belong to one factor.

Multiple factors Multiple between-subjects factors → factorial ANOVA. Between and within factors → split-plot/mixed ANOVA. An interaction means one factor's effect depends on another factor's level.

Limits of inference Correlation r measures association. Causal claims need an appropriate design. Significance, effect size and practical value differ. A p value does not give the probability a hypothesis is true. Check variability, sampling, control, measures and context.

Complementary metrics Success rate $= n_{\text{success}} / n_{\text{attempts}}$. State how assistance was counted. Offset, error rate, time, throughput and satisfaction capture different aspects. Less time/errors or higher throughput can indicate improvement.

Reading studies → conditions and evidence

Noise and mobile interaction Noise type was manipulated. Acquisition, search and text entry used different outcomes (time, offset, error). The 24-person within-subjects study counterbalanced conditions and randomised task order. Effects can vary by task and noise.

MR: movement × load Standing/walking × 0, 0.5, 1.0 kg. Direct selection, ray-casting and text entry. Read MT, offset, error, throughput and UER/CER separately. Test combinations to understand everyday use.

WICE / instrumentation Weight-Induced Consumed Endurance estimates shoulder fatigue from carrying weight. Workload, exertion, physiological strain, endurance time, fatigue and perceived exertion differ. EMG and Borg CR10 provide different evidence. Torque, posture and capacity matter. Camera weight estimation was planned.

Emotion elicitation / sensing A VR vs. 360-degree video study gathered SAM reports of valence and arousal after scenarios. Distinguish stimulus, observed or reported response, and available signal. Responses can differ across people.

Voice assistant recovery Manipulated perceived voice age and response (humorous, apologetic, no acknowledgement). Outcomes included empathy, trust and recovery satisfaction. Interpret combinations in context. Measure actual recovery separately.

Rules for scenario questions

Name the distinction First use / delayed return: learnability / memorability. Missing capability / hard to use: utility / usability. User understanding / design representation: mental / conceptual model. Screen / path / activity: wireframe / wireflow / storyboard.

Match claim to evidence Liking cannot establish completion. Expert predictions need user evidence for confirmation. Logs cannot identify motives alone. Accurate sensing does not establish effective adaptation.

Check scope Read each option's subject, goal, condition and qualifier. Watch for findings applied to all users and planned capabilities described as validated.

Identify data structure first Same people test two interfaces: paired, with order effects to handle. Same people hear three noise conditions: one within-subjects factor, three levels. Noise within and experience between: mixed. Behavioural and self-report measures can coexist.